CRITICAL VALUES FOR PRESERVING FIBONACCI STRUCTURE IN DISK-STACKING MODELS

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ABSTRACT. This is the outline of a paper exploring the computational and hopefully analytic properties of the threshold at which disk-stacking models fail to preserve Fibonacci structure.

1. INTRODUCTION

This note explores the dynamics of Schwendener disk-stacking models reintroduced into phyllotaxis by Douady and Golé. A tutorial introduction to these models, along with further references can be found in my textbook [2]. They are of particular interest, in part because they connect with the known biology of plant organ formation, but more relevantly here because they are capable of showing Fibonacci structure. In particular, they have a geometric parameter which is the ratio of the size of the disks being stacked to the arena in which they are being stacked in, and when this parameter is smoothly and slowly changed, the resulting arrangement of disks exhibits families of spirals characterised by increasing pairs of Fibonacci numbers. If the geometric parameter is changed too quickly, this property is lost, and in [1] I explored with some success whether this, in conjunction with the introduction of noise, could explain observed biological data which often includes non-Fibonacci spiral counts and sometimes the complete loss of spiral structure. That paper reconfirmed the observations of Douady and Golé that there is a non-dimensional threshold for the rate of change of the geometric parameter above which non-Fibonacci transitions occur, but at present this threshold is poorly characterised numerically and not at all analytically.

2. METHODS

Instead of the fixed-circumference cylinder bur changing stacking-disk radius of previous models, here I keep the radius of each stacking disk constant but vary the circumference of the periodic arena in which it is stacked. In the cylindrical model, the disk radius was modelled as $r(z)$, where z was vertical height on the cylinder, and $r(z)$ was in simple cases linear in z. Then $r' = dr/dz$ was a non-dimensional measure of the rate of change of disk size. Here, instead we keep $r(z) == 1$ but vary the cone circumference linearly with z. This has numerical advantages and is perhaps easier to interpret visually, and r' becomes roughly $1/\tan\theta$ where θ is the semi-angle of the cone.

The simulation output depends only on the semi-angle θ and the initial configuration of disks. In every case in this paper, the initial configuration is a single disk placed to fill the lowest part of the cone; since the disks hav fixed radius $r = 1$, if the apex of the cone is at $(0,0)$ this first disk is placed at $(0, 1/2 * 1/\sin(\theta)$.

The simulation code relies on a Wolfram notebook at `https://github.com/js229/GeometricalPhyllotaxis/blob/master/Mathematica/Packages/CoinStacking.nb`, and the simulations themselves are run by `https://github.com/js229/GeometricalPhyllotaxis/`

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`blob/master/LaTeX/Critical%20values/CriticalR.nb`; the first line of the latter notebook needs to point to the location of the first.

Figure 1 gives an example of the simulation.

Parastichy lines are constructed from the finished simulation as explained in the Figure caption, and parastichy counts are calculated as how many different parastichy lines can be seen in at the top rim of the cone where the simulation has been terminated.

3. Results

Figure 2 shows the parastichy counts as a function of the speed at which geometry changes as measured by the semi-angle of the cone. For narrow cones with semi-angle of about 0.155π, each transition follows the Fibonacci rule associated with a pattern locally near a hexagonal packing as discussed in [2]. For values of the cone semi-angle beyond around 0.155π, the reliable Fibonacci transitions break down, although there is evidence of continued structure until about 0.165π.

The region near 0.166 where the cone semiangle is $\pi/6$ shows a coherent structure (Figure 3). On examination of the resulting disk pattern this nay because the expanding cone is very nearly able to accomodate a repeated lattice-like pattern.

I would expect the details of the parastichy counts and disk patterns above transition to be quite dependent on the point at which the pattern is evaluated.

Figure 4 gives an example well past the transition.

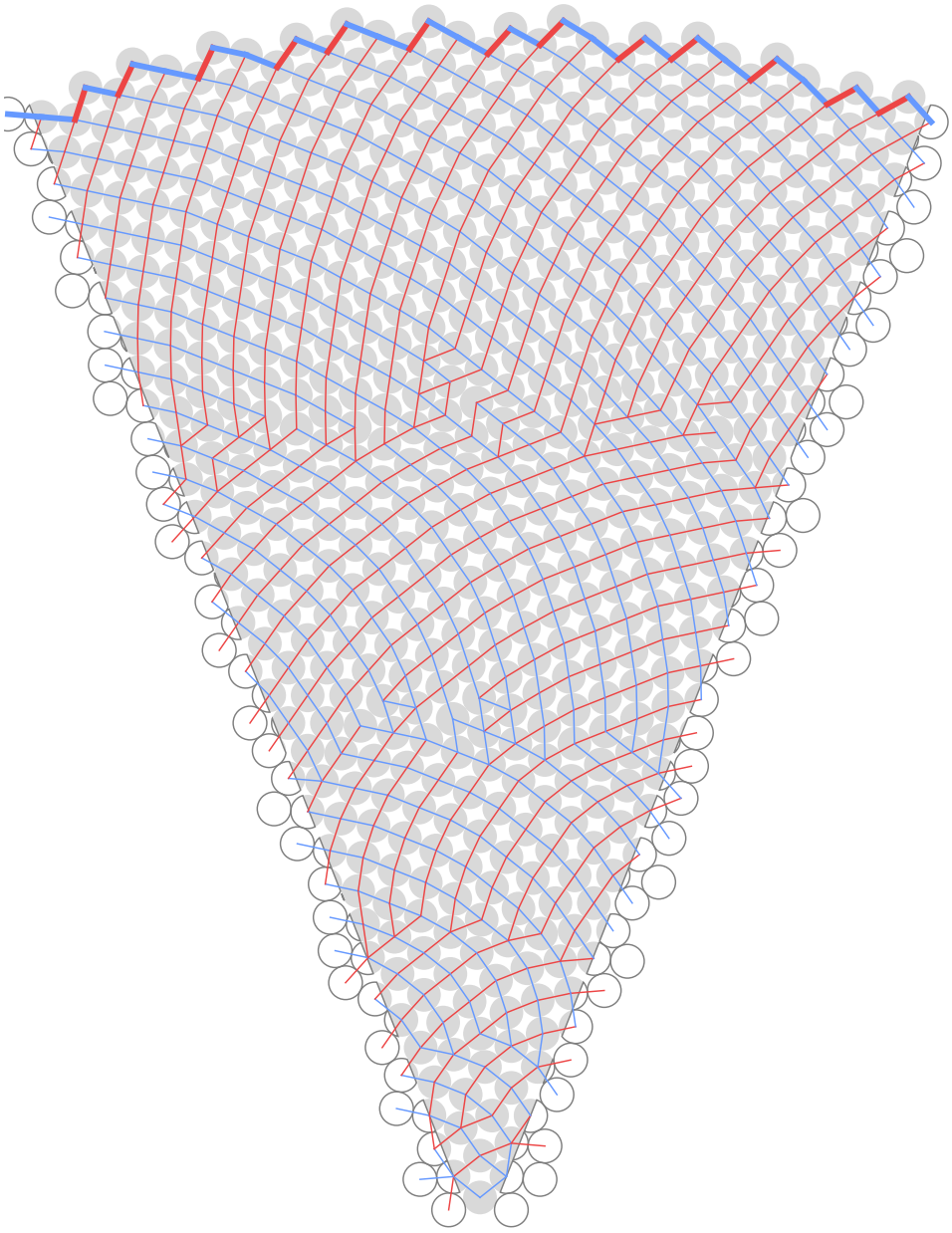

FIGURE 1. A stacked-disk simulation with semi-cone angle 0.12π radians. The disk-stacking takes place on a cone, with periodic copies of disks indicated by the open circles, and at each iteration the next disk is placed at the closest possible position to the cone apex, without overlapping any previously placed disks. Each disk is joined to the two disks which support it from below by a red line to one on its right and a blue line to the one on its left; disks not joined by lines are not touching. This pattern has a $(13, 21)$ parastichy at the top rim, as measured by counting the 13 blue lines running upwards and to the left, taking account of the cone periodicity, and the 21 red lines running upwars and to the right.

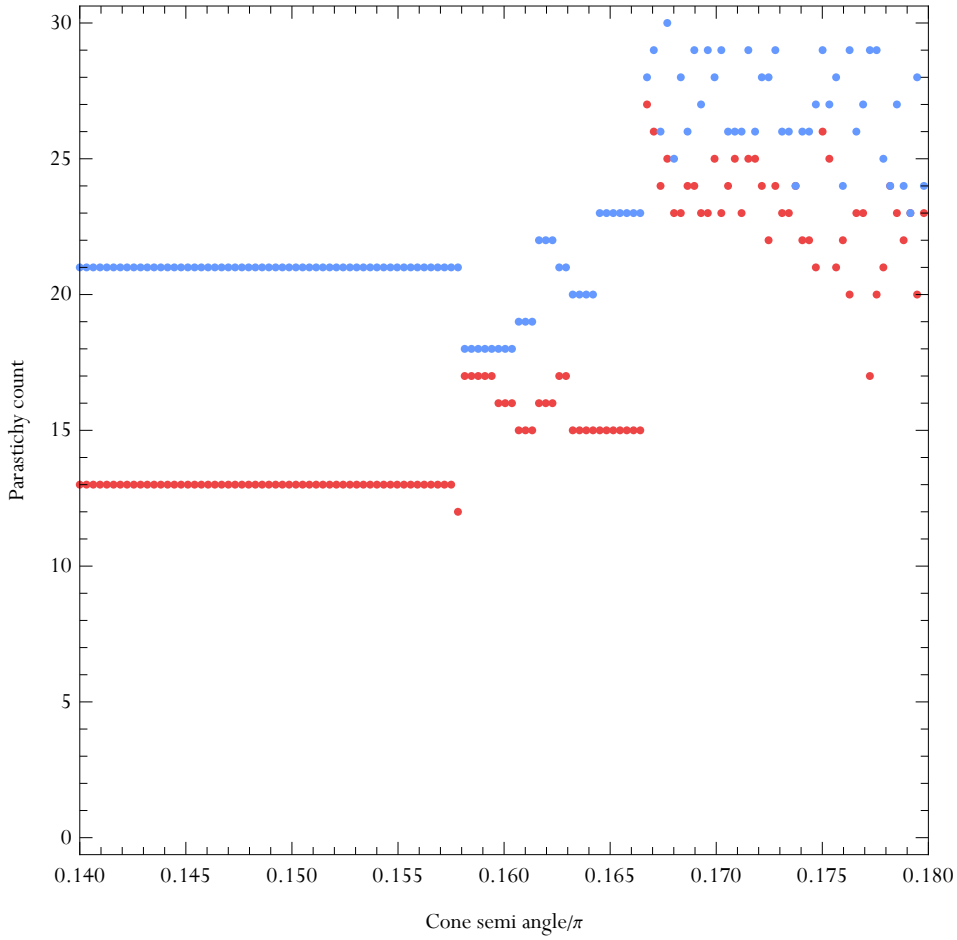

FIGURE 2. First results: parastichy counts as a function of the cone semi-angle. In each case the simulation was run to fill a cone with upper rim circumference $2\pi \cdot 55$

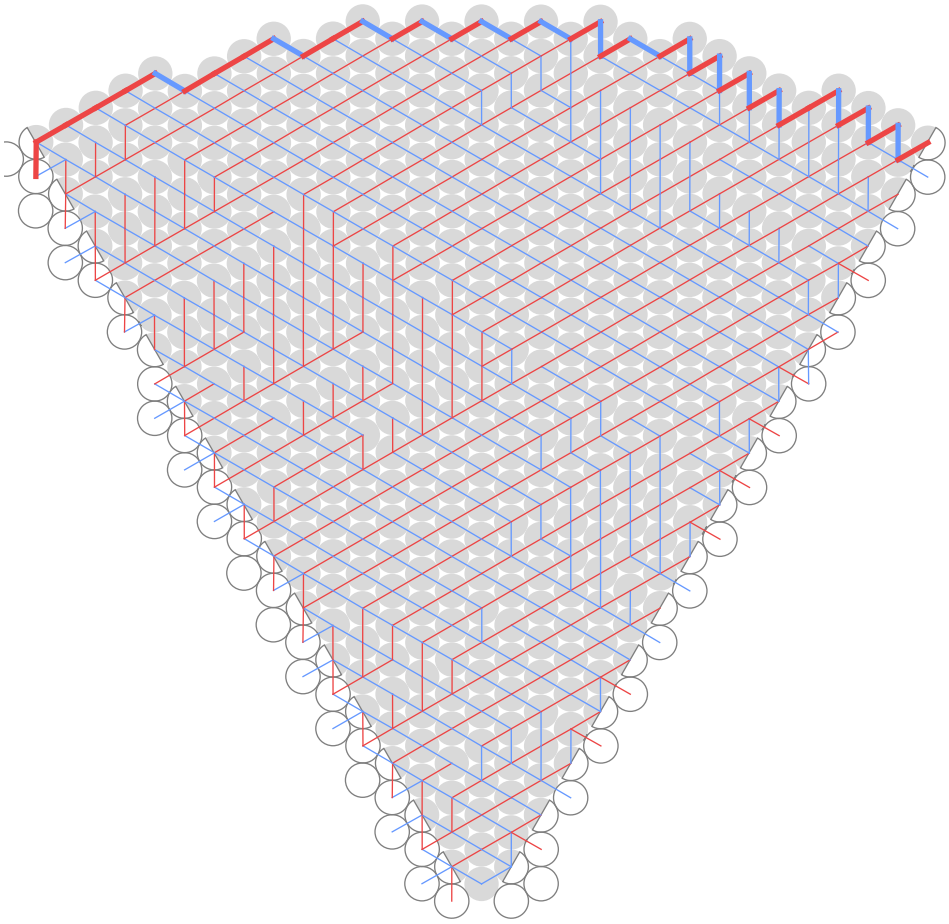

FIGURE 3. Resulting pattern is close to hexagonal when the cone semi-angle is close to $\pi/6$.

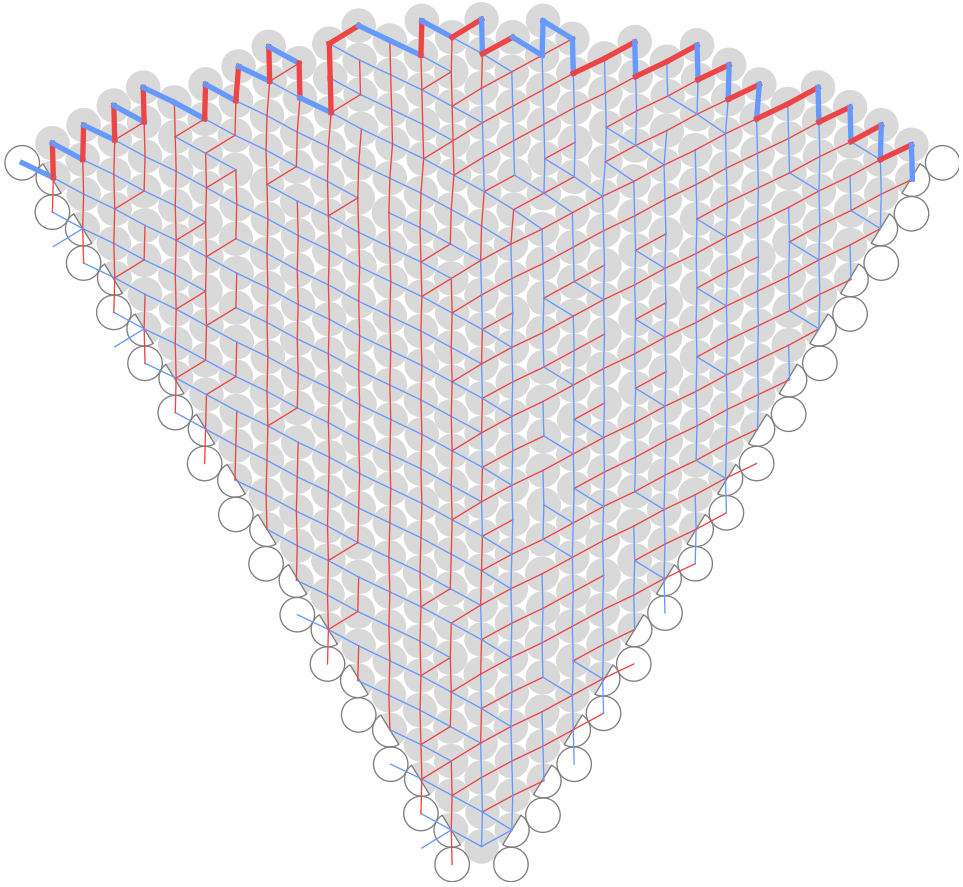

FIGURE 4. Resulting pattern when the cone semi-angle is close to $\pi/5.5$. The vertical parastichy lines reflect the stacked column structure which Doaudy and Golé call quasi-symmetric.

4. Analytic approach

Suppose that we took the chain at each height as coming from the van Iterson touching-circle lattice from the corresponding geometry ratio?

5. Outstanding issues

- Is the threshold cone angle the same size even starting from large Fibonacci patterns
- Make the cone piecewise linear and explore how to reliably generate eg Lucas numbers
- Do the results hold for longer runs
- Add noise along the lines of the previous paper
- fine structure of transition - expect to see lots of islands of structure when we look carefully
- Relate to cylinder paper results and at a minimum connect θ to r'.
- Explore geometry at transition to derive a condition for the critical θ.
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6. Some other topics

Out of scope for this paper, but of interest

- log term dynamics on a cone - Figure 5 looks explicable.
- long term dynamics on a cylinder, ie with no change in relative geometry. This also has lots of good open questions such as whether all attractors are rhombic tilings.

Statements and Declarations

I did not receive support from any organization for this work and have no relevant financial interests to disclose.

References

[1] Jonathan Swinton. *Disk-Stacking Models Are Consistent with Fibonacci and Non-Fibonacci Structure in Sunflowers*. July 2024. arXiv: 2407.05857 [q-bio]. (Visited on 07/09/2024).

[2] Jonathan Swinton. *Mathematical Phyllotaxis*. Springer Nature, 2025. ISBN: 978-3-031-94012-5.

7. Document history

Last compiled April 2, 2026. Mathematica code used to generate the Figures, along with a version-controlled copy of the source text, can be found at `https://github.com/js229/GeometricalPhyllotaxis/tree/master/LaTeX/Critical%20values`.

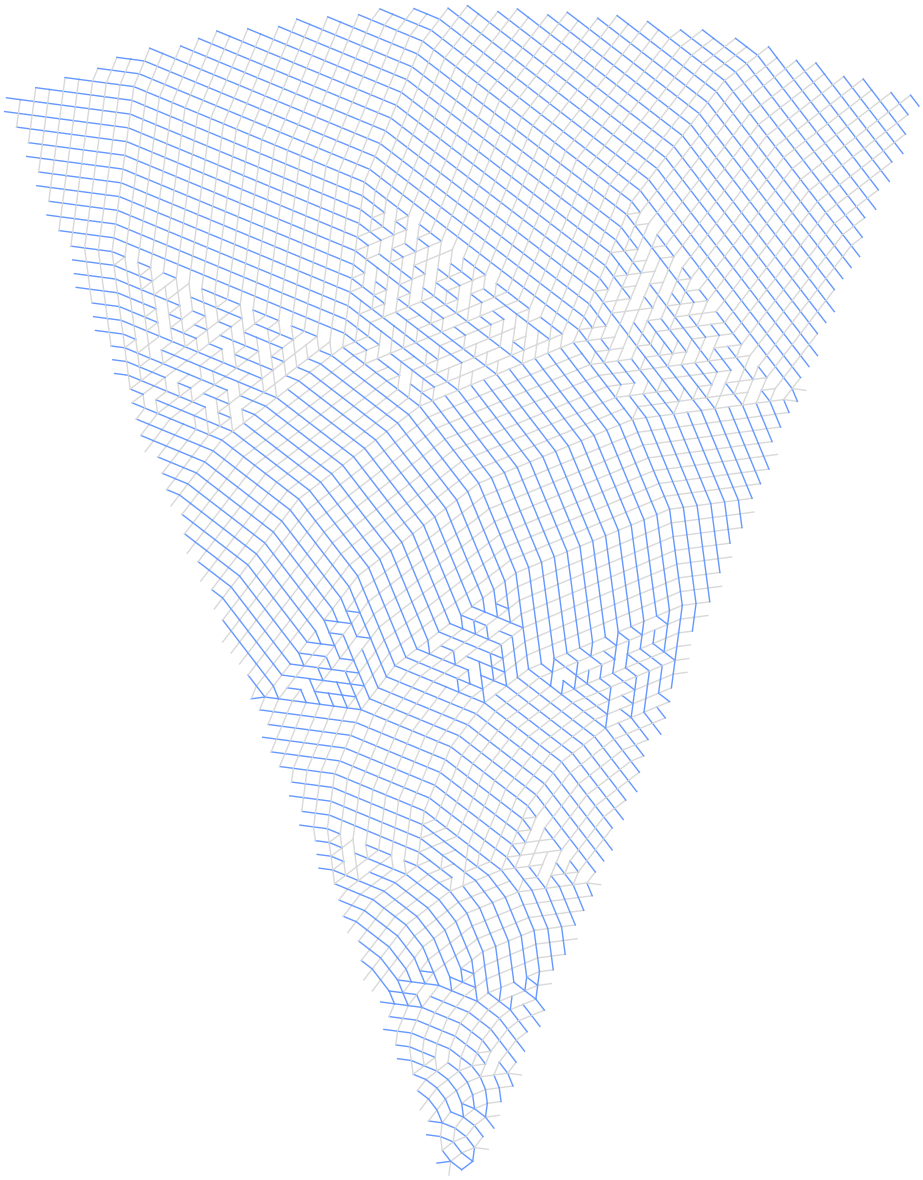

FIGURE 5. Long term Fibonacci transitions have intriguing structure. Diagram as before, but without disks and parastichy lines recoloured to emphasis topological structure.